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[5152]-172

S.E. (I.T.) (I Sem.) EXAMINATION, 2017

COMPUTER ORGANIZATION

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

- N.B. :—**
- (i) Neat diagrams must be drawn wherever necessary.
 - (ii) Figures to the right indicate full marks.
 - (iii) Assume suitable data, if necessary.

1. (a) Perform multiplication operation on the following signed binary numbers using Booth's recording technique. Multiplicand = 0101101, Multiplier = 0011110. Justify whether given multiplier is good or worst. [7]
(b) Draw internal structure of ALU and explain its functioning. [6]
Or
2. (a) Describe function of various signals used in maximum mode operation of 8086. [7]
(b) Perform multiplication operation on the following signed binary numbers using Booth's bit pair recoding technique. Multiplicand = 01101, Multiplier = 1010. Justify whether given multiplier is good or worst. [6]
3. (a) Draw and explain programmer's model of 8086 micro-processor. [6]
(b) Explain the micro operations performed for fetching a word from memory and storing a word in memory. [6]

P.T.O.

Or

4. (a) Explain memory segmentation in 8086. List out the advantages of it. [6]
(b) Compare hardwired control unit with micro-programmed control unit. [6]
5. (a) Explain different memory access methods. [6]
(b) Explain how snoopoy protocol resolves the problem of cache coherence. [6]

Or

6. (a) Explain whether write through or write back policy is better in cache memory. [6]
(b) Write a note on RAID. [6]
7. (a) Explain the need of I/O module. Compare memory mapped I/O with I/O mapped I/O. [7]
(b) Elaborate the feature and functions of 8255. [6]

Or

8. (a) Draw block diagram of USART. [5]
(b) Write a note on peripheral component interface bus. [4]
(c) Explain the significance of HOLD and HLDA signals in DMA operation. [4]